

Notes & Steps



Key idea

The power rule: if $f(x) = x^n$ then $f'(x) = nx^{n-1}$. Multiply by the exponent and reduce the exponent by one.

Power rule table

$f(x)$	$f'(x)$
x^2	$2x$
x^3	$3x^2$
x^4	$4x^3$
x^{10}	$10x^9$
x^n	nx^{n-1}

With a coefficient

If $f(x) = ax^n$ then $f'(x) = anx^{n-1}$. Example:

$$f(x) = 5x^4 \rightarrow f'(x) = 20x^3.$$

Key idea

Differentiation is linear: differentiate term by term and add the results. Constants differentiate to zero.

Sum of terms

$$f(x) = 3x^4 + 6x^3 - 7x^2 - 2x + 3$$

$$f'(x) = 12x^3 + 18x^2 - 14x - 2$$

Fractions as powers

Write as powers of x first:

$$\frac{1}{x} = x^{-1}, \quad \frac{1}{x^2} = x^{-2}, \quad \sqrt{x} = x^{\frac{1}{2}}$$

Then apply the power rule.

Common mistake

Forgetting to multiply by the coefficient too. For $f(x) = 3x^4$, the derivative is $3 \times 4 \times x^3 = 12x^3$, not just $4x^3$.

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Example 1: expand first

Differentiate $f(x) = (2x - 1)(x + 3)$.

Expand: $2x^2 + 6x - x - 3 = 2x^2 + 5x - 3$.

Differentiate: $f'(x) = 4x + 5$.

Example 2: fractions as powers

Differentiate $f(x) = 3x^2 + \frac{2}{x} + 5$.

Rewrite: $3x^2 + 2x^{-1} + 5$.

Differentiate: $f'(x) = 6x - 2x^{-2} = 6x - \frac{2}{x^2}$.

Example 3: roots

Differentiate $f(x) = 4\sqrt{x} - \frac{3}{x^2}$.

Rewrite: $4x^{\frac{1}{2}} - 3x^{-2}$.

Differentiate: $f'(x) = 2x^{-\frac{1}{2}} + 6x^{-3}$.

So $f'(x) = \frac{2}{\sqrt{x}} + \frac{6}{x^3}$.

Example 4: divide through first

Differentiate $f(x) = \frac{4x^3 + 7x^2 - 6x - 5}{x^2}$.

Divide: $4x + 7 - 6x^{-1} - 5x^{-2}$.

Differentiate: $f'(x) = 4 + 6x^{-2} + 10x^{-3}$.

So $f'(x) = 4 + \frac{6}{x^2} + \frac{10}{x^3}$.

Try these

1. $f(x) = 6x^3 - 4x^2 + 2x - 9$

2. $f(x) = (x + 4)(x - 2)$

3. $f(x) = 5\sqrt{x} + \frac{1}{x^3}$

Common mistake

When rewriting fractions, remember each term is divided by the denominator. For $\frac{x^3 + 2x}{x}$, write $x^2 + 2$, not $x^3 + 2$.